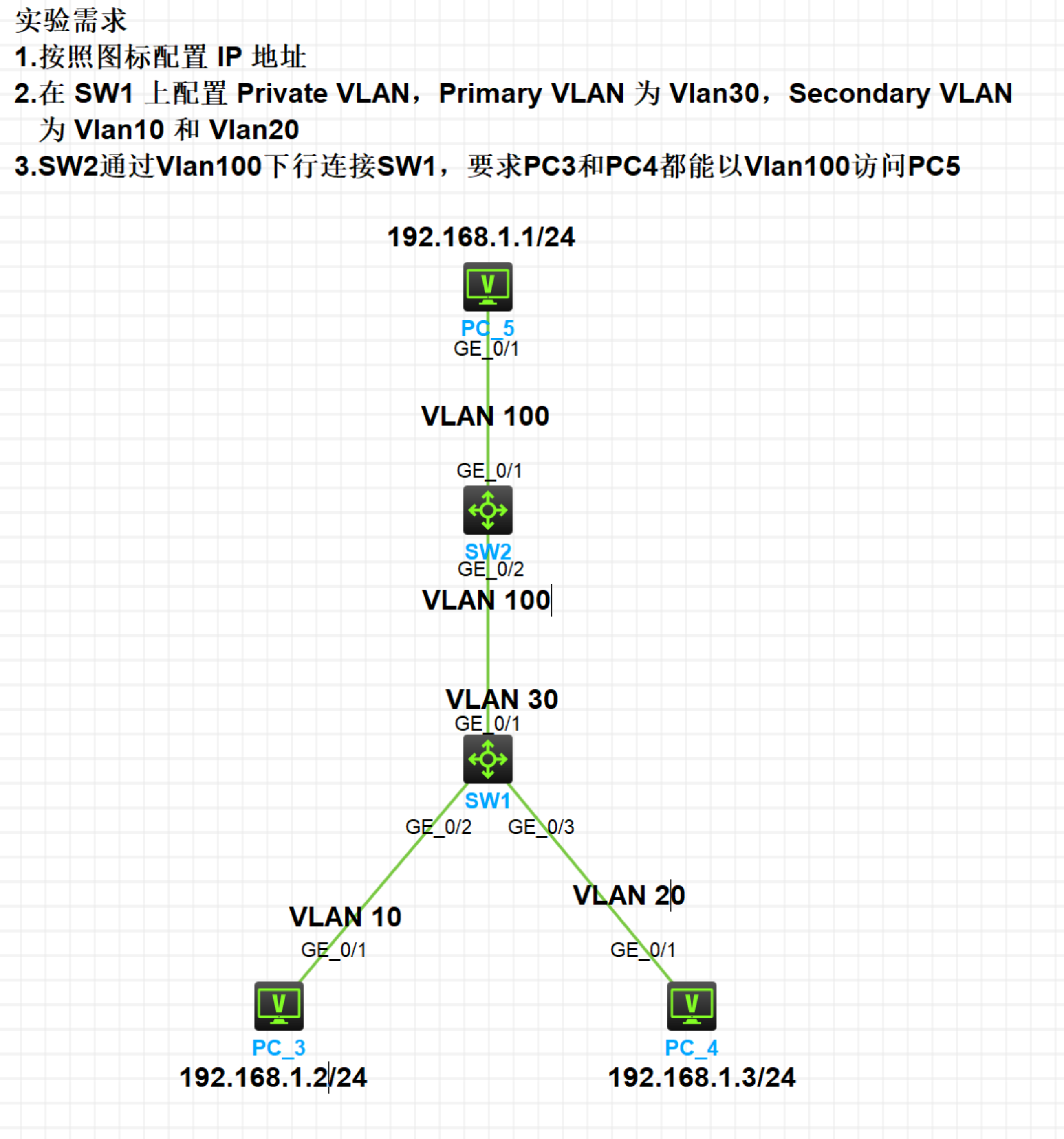
私有VLAN实验

实验拓扑



实验需求

> 1.按照图标配置 IP 地址
>
> 2.在 SW1 上配置 Private VLAN，Primary VLAN 为 Vlan30，Secondary VLAN

> 为 Vlan10 和 Vlan20
>
> 3.SW2通过Vlan100下行连接SW1，要求PC3和PC4都能以Vlan100访问PC5

实验配置

```
[SW2]vlan 100
[SW2-vlan100]port  GigabitEthernet 1/0/1
[SW2-vlan100]port  GigabitEthernet 1/0/2
---
[SW1]vlan 10
[SW1-vlan10]vlan 20
[SW1-vlan20]vlan 30
[SW1-vlan30]private-vlan primary
[SW1-vlan30]%Nov 11 00:01:24:133 2025 SW1 LLDP/5/LLDP_PVID_INCONSISTENT: PVID mismatch discovered on GigabitEthernet1/0/1 (PVID 1), with SW2 GigabitEthernet1/0/2 (PVID 100).
[SW1-vlan30]private-vlan secondary 10 20
[SW1-vlan30]int g1/0/1
[SW1-GigabitEthernet1/0/1]port  private-vlan 30 promiscuous
[SW1-GigabitEthernet1/0/1]int g1/0/2
[SW1-GigabitEthernet1/0/2]port access vlan 10
[SW1-GigabitEthernet1/0/2]port private-vlan host
[SW1-GigabitEthernet1/0/2]int g1/0/3
[SW1-GigabitEthernet1/0/3]port private-vlan host
[SW1-GigabitEthernet1/0/3]port access vlan 20
---

```

结果验证

1.连通性测试

```
<H3C>ping 192.168.1.1
Ping 192.168.1.1 (192.168.1.1): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.1.1: icmp_seq=0 ttl=255 time=1.830 ms
56 bytes from 192.168.1.1: icmp_seq=1 ttl=255 time=1.405 ms
---
<H3C>ping 192.168.1.1
Ping 192.168.1.1 (192.168.1.1): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.1.1: icmp_seq=0 ttl=255 time=1.866 ms
56 bytes from 192.168.1.1: icmp_seq=1 ttl=255 time=1.167 ms
---
<H3C>ping 192.168.1.2
Ping 192.168.1.2 (192.168.1.2): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.1.2: icmp_seq=0 ttl=255 time=1.114 ms
56 bytes from 192.168.1.2: icmp_seq=1 ttl=255 time=1.124 ms


<H3C>ping 192.168.1.3
```

```
Ping 192.168.1.3 (192.168.1.3): 56 data bytes, press CTRL_C to break
56 bytes from 192.168.1.3: icmp_seq=0 ttl=255 time=1.057 ms
56 bytes from 192.168.1.3: icmp_seq=1 ttl=255 time=1.137 ms

```

2.查看Private vlan 信息

```
[SW1]display  private-vlan 30
 Primary VLAN ID: 30
 Secondary VLAN ID: 10, 20

 VLAN ID: 30
 VLAN type: Static
 Private VLAN type: Primary
 Route interface: Not configured
 Description: VLAN 0030
 Name: VLAN 0030
 Tagged ports:   None
 Untagged ports:
    GigabitEthernet1/0/1          GigabitEthernet1/0/2
    GigabitEthernet1/0/3

 VLAN ID: 10
 VLAN type: Static
 Private VLAN type: Secondary
 Route interface: Not configured
 Description: VLAN 0010
 Name: VLAN 0010
 Tagged ports:   None
 Untagged ports:
    GigabitEthernet1/0/1          GigabitEthernet1/0/2

 VLAN ID: 20
 VLAN type: Static
 Private VLAN type: Secondary
 Route interface: Not configured
 Description: VLAN 0020
 Name: VLAN 0020
 Tagged ports:   None
 Untagged ports:
    GigabitEthernet1/0/1          GigabitEthernet1/0/3
---
[SW2-vlan100]display vlan 100
 VLAN ID: 100
 VLAN type: Static
 Route interface: Not configured
 Description: VLAN 0100
 Name: VLAN 0100
 Tagged ports:   None
 Untagged ports:
    GigabitEthernet1/0/1          GigabitEthernet1/0/2
```